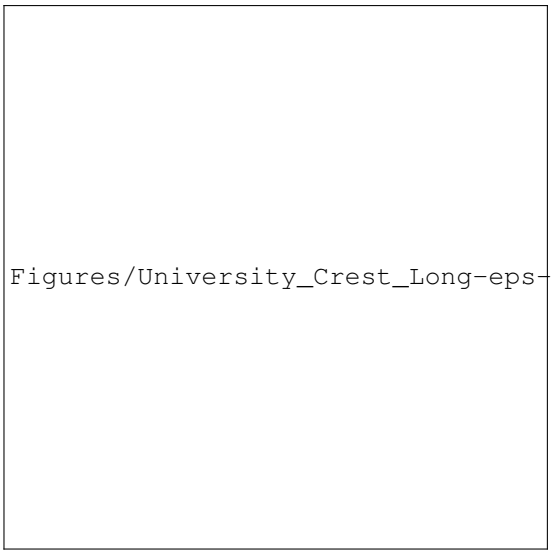
Template Document

Academic Support Agent

February 19, 2025



FIRST YEAR DISSERTATION

submitted towards differentiation for the degree of

Doctor of Philosophy (PhD)

by

School of Mathematics and Physics
Queen's University Belfast

Supervisor(s)

February 19, 2025

*Abstract

Provide an abstract summarising the key information in the report. As a guide, the overall report length should be approximately 10 pages, not including cover page, table of contents or references. The layout of sections provided here is a suggested structure but can be adapted as seen fit by the student and/or supervisor. There are also examples of how to cite references, include figures and cross-reference different parts of the report.

Background In this section, provide a background to the research in the form of a literature review[1] with appropriate citations to relevant work.

You can add both textual and parenthetical references as appropriate — e.g., 2012PhLB..716....1A published an article with very rapid citation rates, but is gradually becoming eclipsed from competition in recent years [e.g.,][to name but a few]2018Natur.556...43C, 2018PhRvD..98c0001T, 2019ApJ...873..111I.

Research to date In this section, please summarise the research undertaken to-date, including figures, tables, initial interpretations of results, etc.. The title can be changed to something more relevant to the work discussed and can be divided into separate sections and subsections as appropriate. A basic layout is below but can be adapted as seen fit.

1 Methodology

Techniques employed to conduct the presented research.

2 Results and Discussion

Figure 1: Plot of $y = x$. Here y represents the really important data and the trend line has been determined using method A.

The original hypothesis is clearly supported by the data shown in figure 1...

3 Conclusions

Summarise the work in this chapter and provide the key scientific conclusions.

Research Plan In this section, please outline the short- and long-term goals of the research including how these will fit into a thesis framework[2]. For example, you can refer the reader to Section 4 for your short-term goals, and Section 5 for your long-term goals.

4 Short-term Goals

My immediate short-term goals are outlined below.

1. Short-term goal 1.
2. Short-term goal 2.
3. Short-term goal 3.

5 Long-term Goals

The project's long-term goals are outlined below.

1. Long-term goal 1.
2. Long-term goal 2.
3. Long-term goal 3.

[1] A literature review may not be suitable for, e.g., pure mathematics students. However, these students should still try to set the scene and explain the need for their research project, i.e., why is it novel and timely to do this research now?

[2] The descriptions of goals and deliverables can be transformed into Gantt charts where applicable